

Independent monotone receivers cannot suppress coincidences

Two independently prepared local receivers, responding monotonically to fixed fractions of one common classical pulse energy, obey $p_{12} \geq p_1 p_2$ on the full gate ensemble. Saturation and unequal efficiencies preserve this constraint. Fluctuating routing, correlated readiness or outcome-dependent gate selection can evade it. This identifies a concrete classical-exclusion premise; the inequality contains no action scale.

Status: Q12 exploratory written derivation and coordinator consistency review. No accepted-ledger promotion. Independent proof review and a bounded sequential librarian comparison are required before promotion.

1. Preparation, local response and counted gates

Fix a gate duration $T > 0$, apparatus settings and pulse shape. In each gate a nonnegative random energy E reaches a fixed passive splitter. Its output energies are aE and bE , where $a, b > 0$ are fixed dimensionless fractions and $a + b \leq 1$. This includes a fixed-phase Q08 interferometer only when its output fractions are fixed across the ensemble. All admitted gates, including empty gates and double records, are counted.

Let $X, Y \in \{0, 1\}$ indicate at least one local record during the gate. The response probabilities are measurable nondecreasing functions

$$f(E) = F_T(aE), \quad g(E) = G_T(bE), \quad 0 \leq f, g \leq 1. \quad (1)$$

Sufficient physical assumptions are independent local apparatus variables U, V , jointly independent of E , and response maps $X = x(aE, U)$, $Y = y(bE, V)$ with the monotone averaged responses (1). There is no communication or shared release during the gate. Mathematically the needed condition is

$$\Pr(X = 1, Y = 1 \mid E) = f(E)g(E). \quad (2)$$

Spatial separation alone does not imply (2). Hidden common readiness, shared reset memory, or an unrecorded pulse shape can invalidate this scalar model. Independence across successive gates is unnecessary for the probability inequality; it would matter for statistical error estimates from finite data.

Define $p_1 = \mathbb{E}X$, $p_2 = \mathbb{E}Y$, and $p_{12} = \mathbb{E}(XY)$. These dimensionless probabilities refer to the same gate ensemble. They are not mean event counts for a receiver allowing multiple records.

2. Exact bound and sharp boundary

For an independent copy E' of E , expand the product to obtain

$$p_{12} - p_1 p_2 = \frac{1}{2} \mathbb{E}[(f(E) - f(E'))(g(E) - g(E'))] \geq 0. \quad (3)$$

Equation (2) gives the equality; common monotonicity gives the sign. Bounded responses make every expectation finite even for energy laws with divergent moments. Thus, if both singles are positive,

$$A := \frac{p_{12}}{p_1 p_2} \geq 1. \quad (4)$$

Equality holds exactly when the nonnegative product in (3) vanishes almost surely. Fixed E gives equality even for unequal detectors. If both responses are strictly increasing on a nondegenerate energy distribution, the inequality is strict. A saturated response may instead give equality.

For example, $F_T(e) = 1 - \exp(-\kappa_1 Te)$ and $G_T(e) = 1 - \exp(-\kappa_2 Te)$ satisfy the bound exactly, with κ_i in $(\text{energy time})^{-1}$. No weak-pulse approximation is required. Independent dark-trigger probabilities d_i replace these responses by $1 - (1 - d_i) \exp(-\kappa_i Te)$ and preserve monotonicity. At fixed pulse energy E_0 the two records are independent and $A=1$. Attenuating E by a factor $\eta^2 \rightarrow 0$ sends the no-dark singles and coincidences to zero while A remains one for every $\eta > 0$. At zero singles A is undefined; (3) remains meaningful.

A practical symmetric target $p_1, p_2 \geq s > 0$ therefore requires $p_{12} \geq s^2$. More generally define the probability of at least one record $q = p_1 + p_2 - p_{12}$ and the channel balance $r = p_1/(p_1 + p_2) \in (0, 1)$. Put $d = p_{12}$ and $c = r(1 - r)$. Then

$$d \geq c(q + d)^2, \quad d \geq \frac{1 - 2cq - \sqrt{1 - 4cq}}{2c}. \quad (5)$$

To see the second statement, move all terms of the first inequality to a quadratic with positive leading coefficient; d must lie between its roots. Here $0 \leq q \leq 1$ and $c \leq 1/4$, so the discriminant is nonnegative. The lower bound is sharp: choose constant independent response probabilities $p_1 = rS$, $p_2 = (1 - r)S$, where $S = (1 - \sqrt{1 - 4cq})/(2c)$. As q runs from zero to one, S runs from zero to $1/\max(r, 1 - r)$, so both probabilities are admissible. For balanced channels this simplifies to

$$d \geq (1 - \sqrt{1 - q})^2. \quad (6)$$

Near-unit event efficiency and balanced channels therefore require frequent doubles in this class. This is a gate-probability trade-off, not an energy gap.

3. Three explicit ways the inference can fail

Fluctuating routing breaks the common ordering. Fix total energy $E_0 > 0$. Let a random routing bit J send the entire pulse to output 1 with probability r and to output 2 otherwise. Use independent deterministic threshold detectors with threshold $0 < \theta < E_0$. Then $p_1 = r$, $p_2 = 1 - r$, $p_{12} = 0$. Local responses to local energy remain monotone; the inputs are no longer fixed fractions of the common scalar E . Conditional on the full routed input, the outputs are deterministic and factorize. Equivalently, opposite bright/dark phases of a two-path network can implement such varying fractions. Mixing those settings is outside (1). This example does not reproduce a fixed-split optical experiment or derive quantum interference statistics.

Correlated readiness breaks conditional independence. Keep both input energies fixed above threshold. Prepare readiness variables $U = J$, $V = 1 - J$ independently of the source and let $X = U$, $Y = V$. Again the singles are r and $1 - r$ with no doubles. Both responses can be monotone in local signal energy, and no communication during the gate is needed. The detectors were prepared with a shared anticorrelated resource. Source independence alone does not exclude this possibility.

Selecting gates by their outcomes breaks the ensemble premise. Start with independent Bernoulli records of equal probability $p \in (0, 1)$. Keep only gates with at least one record. On that selected ensemble,

$$p'_1 = p'_2 = \frac{1}{2 - p}, \quad p'_{12} = \frac{p}{2 - p}, \quad A' = p(2 - p) < 1. \quad (7)$$

The original unselected A equals one. Conditioning on exactly one record makes the apparent coincidence probability zero. Conversely, a source-side herald H is compatible with (3) whenever, within H , (1)–(2) still hold. Correlation of H with E may arbitrarily change the energy distribution; the proof works for that new distribution. A herald that also selects detector readiness needs a new factorization check. Fixed settings should likewise be tested separately; mixtures of oppositely routed settings can create anticorrelation.

4. Decision and the remaining physical obligation

Q11's shared-release model lies outside (2); its exclusive records are therefore consistent with this obstruction. Q12 rules out repairing independent monotone fixed-split receivers merely by changing thresholds, saturation, efficiencies or scalar energy fluctuations. It does not exclude classical models with extra source modes or correlated apparatus preparations. An observed deficit of coincidences identifies a failure of this joint premise package, not uniquely which premise failed.

The statistics are invariant under a change of energy unit and contain no mechanical action observable. Even an independently established violation would require further state and dynamical reconstruction to select a universal positive action parameter. The receiver response constants in section 2 retain their supplied dimensions; no physical clock or Planck constant is inferred.

Park further single-splitter receiver tuning. The next useful quantum-premise test is a bounded two-setting local model: allow arbitrary shared readiness and source variables, but require settings independent of them and local responses with every gate assigned an outcome. Derive the resulting correlation constraint and test a specified singlet preparation against it. This asks whether the shared-preparation escape survives changed measurement settings; it is a new premise test, not an inference of an action scale from Q12. The gap track still needs independently specified physical dynamics before another parent-generator variant can decide its main obligation.

Source comparison and coverage

Grangier, Roger and Aspect, *Europhysics Letters* **1**, 173–179 (1986), primary paper, pp. 174–175, defines gate-normalized singles and coincidences and derives a classical coincidence lower bound using semiclassical detection and an intensity moment inequality. This is an established precedent for the witness, not a novelty claim. The monotone Bernoulli-response argument and countermodels above are derived here under their stated assumptions; the source is not credited with those extensions.

Coverage: one discovery query, one readable primary PDF route and one failed route; extracted text of pp. 174–175 was read. Formula extraction is incomplete and the screenshot tool supplied no inspectable image, so exact equation transcription is not claimed. The source's prose states the coincidence-product bound. No source experimental-data audit, full proof audit or exhaustive literature comparison was performed. Q01's existing B67 companion supplies the operational premise distinction used here.